A Prompt-Guided Transformer Network With Redundancy-Aware Endmember Bundles Considering Spectral Variability for Hyperspectral Unmixing

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***Abstract*—Spectral variability (SV) in complex environments brings a dual challenge to accurate hyperspectral unmixing (HU): large intraclass and small interclass differences. Most deep learning (DL) frameworks treat features uniformly and fail to distinguish highly similar endmembers, which easily confuses abundance estimation. To simultaneously tackle this dual challenge, integrating endmember bundle priors that effectively accommodate intraclass variations into deep networks has emerged as a promising solution. However, current methods face bottlenecks in both the extraction and utilization of endmember bundles. To this end, we propose a novel prompt-guided transformer network (PGT-Net). It accommodates intraclass variations while effectively amplifying subtle interclass differences. First, addressing the redundancy and noise introduced by existing extraction methods that blindly pursue quantity, a density-based redundancy-aware endmember bundle extraction (DR-EBE) algorithm is proposed. It utilizes density-guided clustering and a spatial–spectral screening strategy to extract high-quality, low-redundancy endmember bundle priors. Second, our innovatively designed prompt-guided endmember transformer (PGET) converts these priors into endmember-specific prompt vectors. It performs targeted activation of specific features to capture and amplify subtle interclass differences. Finally, a spatial complementary integration module (SCIM) is designed. It utilizes spatial complementarity to adaptively refine feature boundaries and performs cross-channel interactions to further break information isolation. Extensive experiments on multiple hyperspectral datasets demonstrate the effectiveness and competitive performance of PGT-Net compared with existing unmixing algorithms. The code is publicly available at https://github.com/UPCGIT/PGT-Net.**

***Index Terms*—Deep learning (DL), endmember bundle, hyperspectral unmixing (HU), spectral variability (SV), Transformer.**

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I. Introduction

HYPERSPECTRAL data contain rich spectral information and are highly valuable for land-cover classification [1], [2], agricultural assessment [3], [4], [5], and environmental monitoring [6], [7]. However, their low spatial resolution often produces mixed pixels, which severely limit further applications [8]. Hyperspectral unmixing (HU) is the key method for addressing this limitation. Its core tasks are extracting pure material spectra (endmembers) [9], [10], [11] and estimating their proportions within pixels (abundances) [12], [13], [14].

HU relies on the spectral mixture model [15]. It assumes fixed and unique endmember spectra. However, this assumption is overly idealized in real scenes. It ignores heterogeneity-induced spatiotemporal spectral evolution, known as spectral variability (SV) [16], [17], [18].

Environmental, atmospheric [19], and illumination factors induce widespread endmember variability in hyperspectral images (HSIs), presenting a dual challenge for unmixing. On the one hand, the same material may show diverse spectra across regions. This creates large intraclass differences, making a single spectrum inadequate. On the other hand, different materials can exhibit highly similar properties in specific environments. This causes small interclass differences. Such spectral ambiguity has also been identified as a common challenge in HSI analysis [20], [21]. Therefore, unmixing must address both issues simultaneously. Otherwise, it violates physical constraints, leading to distorted endmembers and biased abundances.

To address significant intraclass spectral differences, endmember bundle methods were proposed [22], [23], [24], [25]. A bundle is a set of spectra representing one endmember class, extracted directly from images without extra parameters. Traditionally, algorithms like automatic endmember bundle extraction (EBE) extract and cluster endmembers from randomly sampled image subsets. However, ignoring spatial information risks missing endmembers in some subsets and extracting mixed pixels. To improve this, an EBE method based on spatial and spectral information (SSEBE) [26]

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introduced joint spatiospectral constraints using pixel purity index (PPI) and homogeneity index (HI). Other advances include collaborative extraction based on multiple objective particle swarm optimization (MOPSO) [23] and the wavelet-based spectral curve endmember extraction (SCEE) algorithm [27]. Yet, these methods often blindly pursue quantity. This introduces extreme redundancy and low-quality noise. Consequently, they fail to accurately capture intraclass variations, while the noise further obscures subtle interclass differences. Furthermore, these traditional methods cannot deeply mine the complex spatiospectral structural information inherent in HSI.

Relying on its powerful deep nonlinear fitting capabilities, deep learning (DL) has emerged as a novel approach to addressing SV in unmixing [28], [29], [30], [31]. It can efficiently and simultaneously extract endmembers and abundances [32], [33], [34], [35]. However, traditional networks mostly assume fixed spectra. They struggle with complex SV in real scenes. To resolve this, subsequent research shifted to new DL models. These models can flexibly adapt to and learn SV.

In particular, studies mainly focused on probabilistic generative models to learn latent statistical distributions of SV. For example, DeepGUn [36] uses variational autoencoders to learn the low-dimensional manifold of endmember variability. However, it requires prior pure pixels and alternating optimization, which limits endmember diversity. To remove this constraint, the probabilistic generative model for hyperspectral unmixing (PGMSU) [37] directly uses observed mixed pixels. Variational inference jointly estimates abundances and generates endmembers. Deep networks' nonlinear mapping decodes latent variables into endmembers, achieving unsupervised, adaptive fitting of variability distributions. Later, to capture spatial correlations, Shi et al. [38] proposed a deep generative model for spatial-spectral unmixing (DGMSSU) with a dynamically learnable adjacency matrix. Meanwhile, the spatial-spectral adaptive fusion network (SSAF-Net) [39] captures complex variability details by adjusting feature fusion ratios. Despite iterating to fit complex spectral distributions and mine spatial contexts, these models essentially rely on latent space data fitting, lacking clear spectral physical meaning. Crucially, high interclass similarity in real scenes causes latent distributions of materials to overlap. This confuses posterior inference, preventing targeted optimization of weak, key variability features.

To overcome probabilistic generative models' poor physical interpretability, later networks used endmember bundles as physical constraints. EGU-Net [40] clustered bundles to facilitate two-stream unmixing, offering physical significance but exhibiting sensitivity to noise. To enhance purity, TANet [41] employed superpixel preprocessing, which addresses variability indirectly through weight-sharing. Subsequently, EACNN [42] incorporated spatial-channel attention to enable complex feature extraction. Recently, the endmember-oriented transformer network (EOT-Net) [43] handled variability via a Transformer's directional subspace projection. While bundles partly accommodate intraclass variability and provide priors, their extraction leaves redundancy. They also struggle to distinguish spectrally similar endmembers in real-world scenes. In addition, material distributions inherently possess spatial correlations. However, existing networks process endmember features independently, ignoring their potential spatial interactions. Finally, despite using attention or projections, feature extraction remains undifferentiated. Networks cannot dynamically amplify weak, discriminative details, easily confusing abundance estimations for similar endmembers.

In summary, due to complex spatiospectral heterogeneity, existing unmixing frameworks face dual challenges in handling SV. Endmember bundles mitigate SV, but increasing their number indiscriminately introduces substantial redundancy and boundary mixing, which in turn worsens unmixing challenges when interclass differences are small. Meanwhile, current deep networks process features globally and without discrimination, so they cannot selectively separate highly similar endmembers or enhance subtle distinguishing features. Related hyperspectral studies have also shown that preserving spectral differences and representative information while reducing redundancy is critical to maintaining discriminative capability [44], [45]. Based on the above analysis, an effective unmixing framework, therefore, requires collaborative processing. It must acquire high-quality, representative bundles to eliminate redundancy and noise at the source. During unmixing, it must accurately capture and activate key discriminative features to accommodate intraclass variability while amplifying interclass differences.

In recent years, prompt learning has achieved remarkable progress in computer vision. Its core idea of using prompt information to guide feature representation offers a promising perspective for fine-grained feature discrimination [46], [47]. Inspired by this, we propose a prompt-guided Transformer network (PGT-Net) for HU. PGT-Net is designed to synergistically address the two primary challenges induced by SV: large intraclass differences and small interclass differences. First, to address significant intraclass variability, a density-based redundancy-aware EBE (DR-EBE) module is designed. This module aims to extract high-quality and representative endmember bundles, thereby mitigating redundancy and noise at the source and fully accommodating the intraclass SV of different endmembers. On this basis, a multibranch processing architecture is further constructed. Second, to address the issue of small interclass differences caused by highly spectrally similar endmembers, we introduce a prompt-guided mechanism. We designed a prompt-guided endmember Transformer (PGET) that performs targeted activation of specific endmember features, thereby capturing and amplifying these subtle interclass differences. Finally, considering the lack of effective interaction when processing endmember features independently, a spatial complementary integration module (SCIM) is devised to facilitate spatial-channel synergistic interaction among different endmember features, which further enhances the unmixing accuracy in complex scenes.

In summary, the main contributions of this article are threefold.

1) A DR-EBE algorithm is proposed. It tackles the large intraclass differences caused by spatial–spectral heterogeneity. By integrating density-guided clustering with a spatial–spectral screening strategy, it extracts highly rep-

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resentative, low-redundancy endmember bundles. This removes noise at the source and provides high-quality prior guidance for the subsequent unmixing network.

2) A PGET is designed. It addresses the small interclass differences caused by highly spectrally similar endmembers. This module introduces a prompt-guided mechanism to convert endmember bundle priors into endmember-specific prompt vectors. It overcomes the limitations of traditional uniform global feature extraction. PGET performs targeted activation of specific endmember features from a shared backbone. By simultaneously capturing global context and amplifying weak interclass differences, it greatly enhances the discrimination of similar endmembers.
3) A SCIM is designed to resolve the independent processing and lack of interaction of endmember features in complex scenes. It introduces a dual-stage feature enhancement mechanism. Spatial complementarity adaptively refines feature boundaries, and channel shuffling breaks information barriers among features. This enables effective spatial-channel synergistic interaction. It further guarantees high-precision unmixing in complex environments.

II. Related Works

A. LMM and Endmember Bundle

1) Linear Mixing Model: The linear mixing model (LMM) is a basic premise of HU. It assumes a mixed pixel's observed spectrum is a linear weighted combination of pure endmembers and their abundances. Let $\mathbf{Y} \in \mathbb{R}^{L\times N}$ denote a HSI with L bands and N pixels. The matrix form of the LMM is given by

$$\mathbf{Y} = \mathbf{M}\mathbf{A} + \mathbf{R} \tag{1}$$

$$\text{s.t.} \quad \mathbf{1}^{\mathrm{T}}\mathbf{A} = \mathbf{1}^{\mathrm{T}}, \mathbf{A} \geq 0 \tag{2}$$

where $\mathbf{M} = [\mathbf{m}_1, \ldots, \mathbf{m}_P] \in \mathbb{R}^{L\times P}$ is the spectral matrix of P endmembers. $\mathbf{A} \in \mathbb{R}^{P\times N}$ is the abundance matrix. $\mathbf{R}$ represents noise. Abundance $\mathbf{A}$ typically satisfies the abundance nonnegativity constraint (ANC) and abundance sum-to-one constraint (ASC). The LMM is physically intuitive. However, it assumes each material has a single, fixed spectrum $\mathbf{m}_i$. It ignores SV caused by complex lighting and environments. This causes significant reconstruction errors in real scenes.

2) Endmember Bundle Model: An endmember bundle-based mixing model is proposed to address SV. Unlike fixed endmember models, it represents each class using a bundle of multiple spectral curves. This covers the spectral variation space. In particular, the endmember bundle for the ith class is matrix $\mathbf{B}_i \in \mathbb{R}^{L\times K}$. K is the number of candidate spectra. During unmixing, the ith endmember $\mathbf{m}_{n,i}$ at pixel n is not fixed. It is a linear combination of bundle spectra using intraclass weights

$$\mathbf{m}_{n,i} = \mathbf{B}_i\mathbf{w}_{n,i} \tag{3}$$

where $\mathbf{w}_{n,i} \in \mathbb{R}^{K\times 1}$ is the intraclass weight of pixel n for the ith bundle. It reflects specific local spectral features. Substituting this into the LMM yields the bundle-based mixing model

$$\mathbf{Y} = \sum_{i=1}^{P} (\mathbf{B}_i\mathbf{S}_i) \odot \mathbf{A}_i + \mathbf{R} \tag{4}$$

where $\mathbf{S}_i = [\mathbf{w}_{1,i}, \ldots, \mathbf{w}_{N,i}] \in \mathbb{R}^{K\times N}$ contains the intraclass weights of all pixels for the ith endmember. $\mathbf{A}_i$ is the expanded abundance matrix. Therefore, the unmixing accuracy is jointly determined by the coverage of $\mathbf{B}_i$ over real spectral variations and the accuracy of $\mathbf{S}_i$ in reflecting local illumination and material properties.

B. Efficient Vision Transformers

Recently, Transformers have been introduced to HU due to their excellent global context modeling. Unlike CNNs with limited local receptive fields, Transformers capture long-range spectral–spatial dependencies. This utilizes global image information more effectively for unmixing. However, standard multihead self-attention (MHSA) faces a major computational bottleneck. Its computation is based on scaled dot-product attention

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathrm{T}}}{\sqrt{D}}\right)\mathbf{V} \tag{5}$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N\times D}$. This complexity grows quadratically with the number of pixels N, i.e., $O(N^2)$. HSIs usually contain massive pixels. Thus, standard MHSA causes huge memory usage and slow inference. To solve this, linear attention was proposed. It introduces a kernel function $\phi(\cdot)$ and changes the computation order via matrix associativity

$$\text{LinearAttention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \phi(\mathbf{Q})(\phi(\mathbf{K})^{\mathrm{T}}\mathbf{V}). \tag{6}$$

This reduces the complexity to $O(N)$, which is linear in the number of pixels. It provides a mathematical basis for building efficient unmixing networks.

III. Proposed Method

Fig. 1 presents the overall architecture of the proposed PGT-Net. At the input stage, DR-EBE applies density-guided clustering and spatial–spectral screening to the HSI to extract high-quality, low-redundancy endmember bundles that serve as reliable endmember priors. These bundles, together with image features, are then fed into a multibranch backbone. Within that backbone, the PGET module transforms the priors into endmember-specific prompt vectors for targeted activation of corresponding endmember features. The resulting branch-wise features are further enhanced and integrated by SCIM through spatial complementary refinement and channel interaction, alleviating boundary ambiguity and promoting interchannel information exchange. The SCIM-based abundance branch then estimates the abundance maps. A bundle-based endmember generation module further predicts pixel-wise endmembers, which are combined with the estimated abundances for spectral reconstruction.

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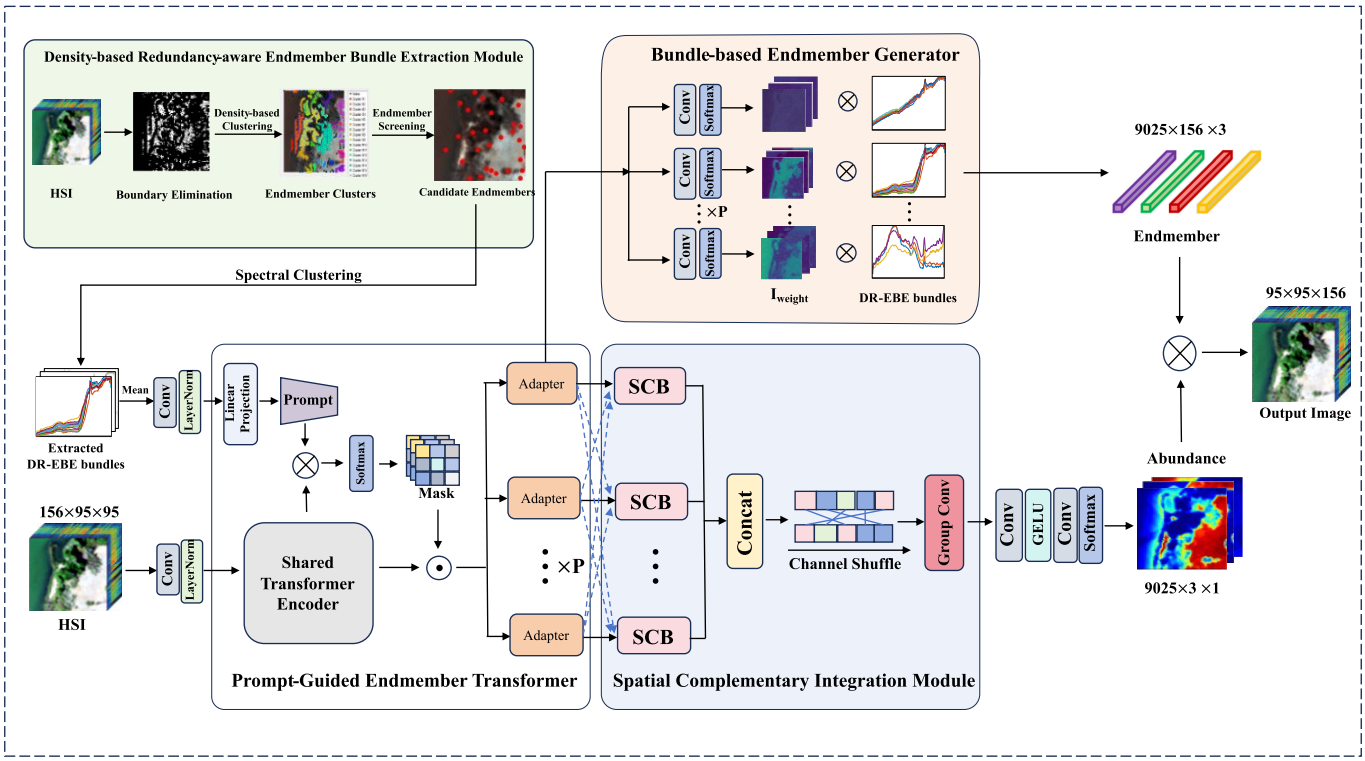


Fig. 1. Overall architecture of PGT-Net.

A. *Density-based Redundancy-aware Endmember Bundle Extraction*

To address large intraclass SV, we designed the DR-EBE module, as shown in Fig. 2. It considers endmember completeness and redundancy. It also uses spatial–spectral information to screen endmembers. Given an input HSI $\mathbf{Y} \in \mathbb{R}^{L\times N}$, it involves three core steps.

1) *Endmember Extraction and Boundary Removal:* To construct a more comprehensive candidate endmember set, we first divide the image into patches and apply the PPI algorithm to each patch. Its score-based pure-pixel detection retains multiple high-purity candidates before redundancy removal. PPI repeatedly projects pixels onto random vectors and counts how many times each pixel is identified as a projection extremum. Pixels that appear more frequently as maxima or minima are assigned higher PPI scores. The candidate set is then defined as $\Omega_{\text{PPI}} = \{\mathbf{y}_i \,|\, \text{PPI}(\mathbf{y}_i) > T\}$, where T is a predefined purity threshold. Since boundary pixels at feature junctions are highly likely to be mixed pixels, we exclude them. In particular, we extract the first principal component of $\mathbf{Y}$ and use the Canny edge detector to identify boundaries. By marking their four-neighborhood positions, we obtain the purified, high-confidence candidate endmember set $\varepsilon_{\text{cand}}$.
2) *Density Clustering and Selection:* To eliminate spatial redundancy in homogeneous regions, we use the density-based clustering algorithm (DBSCAN) to adaptively cluster candidate endmembers according to their spatial density, as it can handle irregularly distributed candidates without predefining the number of clusters. By expanding clusters from core points with sufficiently dense neighborhoods, DBSCAN groups spatially aggregated candidates, while isolated candidates are regarded as noise and removed, yielding several spatial endmember clusters. To select a representative for the ith cluster, we first calculate its spatial geometric center

$$\mathrm{G}^i = \left(\frac{1}{N^i} \sum_{k=1}^{N^i} x_k^i, \frac{1}{N^i} \sum_{k=1}^{N^i} y_k^i \right). \tag{7}$$

Next, to select a unique, representative endmember for each cluster, we design a joint spatial–spectral evaluation function

$$J_k^i = \lambda_1 U_k^i + \lambda_2 D_k^i \tag{8}$$

where U_k^i is the PPI score of the kth candidate in cluster i. D_k^i is its spatial distance to the center G^i, and λ_1 and λ_2 are weight parameters. In particular, λ_2 is set to a negative value ($\lambda_2 < 0$), so that candidates located farther from the cluster center are penalized. By balancing spectral purity and spatial representativeness, the candidate with the highest score becomes the support endmember $\mathbf{m}^i$

$$\mathbf{m}^i = \text{argmax}_k \left\{ J_k^i \right\}_{k=1}^{N^i} \tag{9}$$

where J_k^i is the weighted evaluation value and N^i is the total number of candidates in cluster i. This stage compresses the redundant candidate set into a support endmember set $\mathbf{M}$ with clear physical meaning.
3) *Endmember Spectral Clustering:* After the above screening, the support endmembers retain representative

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spectral variations but may belong to different endmember classes. Since the previous steps have reduced spatial redundancy and unreliable candidates, the remaining task is to group these support endmembers according to spectral similarity. Therefore, inspired by the spectral clustering strategy used in SSEBE, we apply K-means clustering to group the selected support endmembers into bundles according to spectral similarity. In particular, K-means finds P spectral centers that minimize the distance between each endmember vector and its nearest center, where P is the number of endmembers. The resulting bundles fully capture endmember SV with extremely low intraclass redundancy, providing stable prior guidance for subsequent abundance estimation.

B. Prompt-Guided Endmember Transformer

To overcome small interclass differences, we design a PGET module that employs endmember prompt guidance to capture and amplify weak discriminative features. It involves three core steps.

1) *Shared Global Feature Modeling:* Since unmixing requires ample contextual information to handle complex SV, we first extract joint spatial–spectral features under a global receptive field. Given an input HSI $\mathbf{Y} \in \mathbb{R}^{L\times H\times W}$ (where $N = H \times W$), the network performs shallow mapping via convolution and layer normalization (LN). Combined with patch embedding, the input sequence $\mathbf{Z}_0$ is constructed as follows:

$$\mathbf{Z}_0 = \mathrm{LN}\left(\Phi\left(\mathbf{Y}\right)\right) + \mathbf{E}_{\mathrm{pos}} \in \mathbb{R}^{N\times D} \tag{10}$$

where $\mathbf{E}_{\mathrm{pos}}$ is a learnable spatial positional encoding used to preserve pixel topological relationships. $\mathbf{Z}_0$ is fed into the shared Transformer encoder. The operations in each layer are

$$\begin{cases} \mathbf{Z}'_l = \mathrm{MHSA}\left(\mathrm{LN}\left(\mathbf{Z}_{l-1}\right)\right) + \mathbf{Z}_{l-1} \\ \mathbf{Z}_l = \mathrm{MLP}\left(\mathrm{LN}\left(\mathbf{Z}'_l\right)\right) + \mathbf{Z}'_l \end{cases} \tag{11}$$

where MHSA denotes multihead linear self-attention and MLP is a multilayer perceptron. In each encoder layer, we adopt the linear attention mechanism introduced in Section II-B to ensure high computational efficiency under a full-image input strategy. In particular, the attention queries and keys are kernel-mapped to reformulate the attention computation as $\phi(\mathbf{Q})(\phi(\mathbf{K})^{\mathrm{T}}\mathbf{V})$, enabling the shared Transformer encoder to capture long-range spatial–spectral dependencies with $O(N)$ complexity. The final layer's output is reshaped into a shared feature map $\mathbf{F}_{\mathrm{shared}} \in \mathbb{R}^{D\times H\times W}$.

2) *Prompt-Guided Activation:* PGET also receives the mean endmember bundle spectra $\bar{\mathbf{b}}_i \in \mathbb{R}^L$ from DR-EBE as physical priors. To align the physical and semantic domains, a projection function $\Psi(\cdot)$ converts them into endmember-specific prompt vectors $\mathbf{p}_i$

$$\mathbf{p}_i = \mathrm{Linear}(\mathrm{LN}(\mathrm{Conv}(\bar{\mathbf{b}}_i))) \in \mathbb{R}^D. \tag{12}$$

These prompt vectors contain highly concentrated, class-specific discriminative features and serve as endmember prompts for subsequent feature filtering.

During feature interaction, we compute the scaled dot product between the shared feature map $\mathbf{F}_{\mathrm{shared}}$ and $\mathbf{p}_i$ to evaluate semantic similarity. To suppress background noise and sharpen target features, we introduce a sharpening factor γ before the softmax activation to generate an endmember-specific probability attention mask $\boldsymbol{\eta}_i$

$$\boldsymbol{\eta}_i = \mathrm{Softmax}\left(\gamma \cdot \frac{\mathbf{F}_{\mathrm{shared}}\mathbf{p}_i^{\mathrm{T}}}{\sqrt{D}}\right) \tag{13}$$

where $\sqrt{D}$ is the dimension scaling factor. The softmax normalizes scores along the endmember category dimension. The mask is then element-wise multiplied with the shared feature map

$$\mathbf{F}^i_{\mathrm{guided}} = \mathbf{F}_{\mathrm{shared}} \odot \boldsymbol{\eta}_i. \tag{14}$$

The factor γ is a fixed empirical parameter and is set to 2.0 in all experiments. It increases the variance of attention weights, making the mask's distribution sparser. Such polarized weight allocation actively filters out interference from spectrally similar nontarget endmembers while strongly amplifying the weak interclass differences selected by the prompt vector, thereby ensuring the precise extraction of highly spectrally similar endmembers.

3) *Multibranch Adaptive Mapping:* Although the target-activated features initially highlight the exclusive traits of target endmembers, residual entanglement and mutual interference inevitably remain because they are all extracted from a shared backbone network. To address this, the activated features are fed into P parallel adapter branches for deep nonlinear separation, with each branch executing the following mapping logic:

$$\mathbf{F}^i_{\mathrm{out}} = \mathbf{W}_2\mathrm{GELU}\left(\mathbf{W}_1\mathrm{LN}\left(\mathbf{F}^i_{\mathrm{guided}}\right)\right). \tag{15}$$

This operation effectively accomplishes complete feature separation. The final P sets of independent feature representations not only fully accommodate intraclass variability but also provide feature inputs with extremely high class purity for the subsequent SCIM module. For simplicity, the output of the ith adapter branch is denoted as $\mathbf{F}_i = \mathbf{F}^i_{\mathrm{out}}$. The resulting endmember-specific features $\{\mathbf{F}_i\}_{i=1}^P$ are split into two parallel paths: one is fed into SCIM for spatial-channel feature integration and subsequent abundance estimation, while the other is directly fed into the bundle-based endmember generator to produce spatially varying endmembers.

C. Spatial Complementary Integration Module

Material distributions exhibit spatial correlation and complementarity. To exploit this relationship, we design the SCIM module (Fig. 3). SCIM performs spatial and channel interactions among branch-wise endmember features to reduce confusion between spectrally similar endmembers. It consists of two stages, namely spatial complementary fusion and channel interactive fusion. The former adaptively incorporates complementary features from other branches to refine

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ambiguous feature boundaries, while the latter promotes interaction among endmember features. Together, they enhance the network's perception of subtle differences and enable accurate abundance estimation.

1) *Spatial Complementary Fusion:* For P parallel endmember branches, let $\mathbf{F}_i \in \mathbb{R}^{D\times H\times W}$ denote the input feature of the ith branch ($i = 1, 2, \ldots, P$). SCIM first performs local spatial correction for each branch through a spatial complementary block (SCB). To remove irrelevant interference before fusing neighborhood features, SCB extracts the filtered feature $\mathbf{H}_i$ of the current target endmember. Meanwhile, since a single feature struggles to distinguish highly similar endmembers, the network aggregates outputs from all other branches as the neighborhood complementary feature $\mathbf{C}_i$, introducing spatial background context

$$\mathbf{H}_i = \sigma\left(\mathcal{F}_{\text{conv}}\left(\mathbf{F}_i\right)\right) \odot \mathbf{F}_i \tag{16}$$

$$\mathbf{C}_i = \sum_{j=1, j\neq i}^{P} \left(\sigma\left(\mathcal{F}_{\text{conv}}\left(\mathbf{F}_j\right)\right) \odot \mathbf{F}_j\right) \tag{17}$$

where $\sigma(\cdot)$ denotes the sigmoid activation function. $\mathcal{F}_{\text{conv}}$ represents a 1×1 convolution. $\odot$ denotes element-wise multiplication. However, blindly introducing global spatial information causes severe redundant noise. Therefore, we designed a dynamic gating network with a channel reduction structure. Acting as an information valve, it adaptively predicts the spatial weight matrix $\mathbf{\Gamma}_i \in \mathbb{R}^{1\times H\times W}$ to dynamically control the injection ratio of complementary features

$$\mathbf{\Gamma}_i = \sigma\left(\mathcal{F}_{\text{c2}}\left(\delta\left(\mathcal{F}_{\text{c1}}\left(\mathbf{F}_i\right)\right)\right)\right) \tag{18}$$

where $\delta(\cdot)$ is the ReLU activation function, while $\mathcal{F}_{\text{c1}}$ and $\mathcal{F}_{\text{c2}}$ represent dimensionality reduction and expansion convolutions, respectively. Finally, we derive the enhanced feature $\tilde{\mathbf{F}}_i$ via residual connection fusion to ensure deep network gradient stability

$$\tilde{\mathbf{F}}_i = \mathbf{F}_i + \mathbf{H}_i + \mathbf{\Gamma}_i \odot \mathbf{C}_i. \tag{19}$$

Equation (19) encourages the model to exploit spatial exclusion and co-occurrence relationships among different endmember responses. The $\mathbf{\Gamma}_i \odot \mathbf{C}_i$ term effectively sharpens and compensates the target endmember's response boundary, filtering out interference from similar spectra.

2) *Channel Interactive Fusion:* Although SCB achieves spatial enhancement, endmember features remain isolated in the channel dimension. To break this barrier, we concatenate P enhanced features along the channel dimension. This yields $\mathbf{F}_{\text{cat}} \in \mathbb{R}^{(P\times D)\times H\times W}$. We then apply a channel shuffle operation for interleaved recombination. This allows the GELU-activated group convolution to extract joint features $\mathbf{F}_{\text{fuse}} \in \mathbb{R}^{D\times H\times W}$ across independent branches

$$\mathbf{F}_{\text{fuse}} = \text{GELU}\left(\mathcal{F}_{\text{group}}\left(\text{Shuffle}\left(\mathbf{F}_{\text{cat}}\right)\right)\right). \tag{20}$$

This cross-channel recombination helps the model reallocate relative endmember abundances globally. Subsequently, the fused features enter the abundance decoding stage. Considering the continuous distribution of real objects, features first pass through a 5×5 convolution to capture local spatial smoothness priors. A 1×1 convolution then precisely compresses them to the endmember category dimension. Finally, a softmax activation outputs the final abundance map $\hat{\mathbf{A}} \in \mathbb{R}^{P\times H\times W}$

$$\hat{\mathbf{A}} = \text{Softmax}\left(\mathcal{F}_{1\times 1}\left(\text{GELU}\left(\mathcal{F}_{5\times 5}\left(\mathbf{F}_{\text{fuse}}\right)\right)\right)\right). \tag{21}$$

The softmax activation enforces the ANC and ASC. The resulting abundance map is then reshaped into the abundance matrix $\hat{\mathbf{A}} = [\hat{\mathbf{a}}_1, \hat{\mathbf{a}}_2, \ldots, \hat{\mathbf{a}}_N] \in \mathbb{R}^{P\times N}$, where $\hat{\mathbf{a}}_n \in \mathbb{R}^P$ denotes the abundance vector of the nth pixel. The estimated abundances are subsequently combined with the spatially varying endmembers generated by the parallel bundle-based endmember generation branch for image reconstruction.

D. Bundle-Based Endmember Generator

To model intraclass SV, the bundle-based endmember generator uses the endmember-specific features produced by PGET to predict pixel-wise intrabundle combination weights over the candidate spectra extracted by DR-EBE. In this way, the DR-EBE bundles serve as fixed physical priors, while the generator learns pixel-adaptive endmember spectra.

For the ith endmember branch, let $\mathbf{F}_i \in \mathbb{R}^{D\times H\times W}$ denote the endmember-specific feature, and let $\mathbf{B}_i \in \mathbb{R}^{L\times K_i}$ denote the corresponding endmember bundle, where K_i is the number of candidate spectra in the ith bundle. The intrabundle weight matrix is obtained through a 1×1 convolution and softmax normalization

$$\begin{aligned} &\mathbf{S}_i = \text{Reshape}\left[\text{Softmax}_{K_i}\left(\mathcal{F}^i_{\text{conv}}\left(\mathbf{F}_i\right)\right)\right] = \left[\mathbf{w}_{1,i}, \ldots, \mathbf{w}_{N,i}\right] \\ &\mathbf{w}_{n,i} \geq 0, 1^{\text{T}}\mathbf{w}_{n,i} = 1. \end{aligned} \tag{22}$$

The spectrum of the ith endmember at pixel n is generated as follows:

$$\mathbf{m}_{n,i} = \mathbf{B}_i\mathbf{w}_{n,i} \in \mathbb{R}^L. \tag{23}$$

Then, the pixel-wise endmember matrix is constructed as follows:

$$\mathbf{M}_n = \left[\mathbf{m}_{n,1}, \ldots, \mathbf{m}_{n,P}\right] \in \mathbb{R}^{L\times P}. \tag{24}$$

Given the estimated abundance vector $\hat{\mathbf{a}}_n$, the reconstructed spectrum is expressed as follows:

$$\hat{\mathbf{y}}_n = \mathbf{M}_n\hat{\mathbf{a}}_n, \quad \hat{\mathbf{Y}} = \left[\hat{\mathbf{y}}_1, \ldots, \hat{\mathbf{y}}_N\right] \in \mathbb{R}^{L\times N}. \tag{25}$$

E. Loss Function

To guide stable network convergence and evaluate unmixing performance, we combine mean squared error (MSE) and spectral angle distance (SAD) into an overall objective function. This constrains the network via spectral amplitude approximation and geometric shape preservation.

1) *MSE Loss:* MSE minimizes the absolute numerical difference between the real observation $\mathbf{Y}$ and the reconstruction $\hat{\mathbf{Y}}$. This constrains the overall spectral

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amplitude, ensuring the reconstructed image numerically matches the true observation

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{\mathbf{Y}}_i - \mathbf{Y}_i\right)^2 \tag{26}$$

2) *Spectral Fidelity Loss:* Relying solely on numerical constraints introduces bias due to amplitude scaling interference from SV. Thus, we introduce SAD to calculate the high-dimensional angle between real and reconstructed spectra. It is insensitive to amplitude changes and effectively preserves the spectral curve's geometric shape

$$\mathcal{L}_{\text{SAD}} = \frac{1}{N}\sum_{i=1}^{N}\arccos\left(\frac{\langle \mathbf{Y}_i, \hat{\mathbf{Y}}_i\rangle}{\|\mathbf{Y}_i\|_2\|\hat{\mathbf{Y}}_i\|_2}\right). \tag{27}$$

3) *Total Loss:* To adapt to varying noise levels and variability across datasets, we introduce nonnegative hyperparameters α and β to flexibly balance the optimization weights of the above two terms

$$\mathcal{L}_{\text{total}} = \alpha\mathcal{L}_{\text{MSE}} + \beta\mathcal{L}_{\text{SAD}}. \tag{28}$$

IV. Experiments

To evaluate model performance and verify module effectiveness, we conducted comparative and ablation experiments on multiple datasets. We compared our model against several current state-of-the-art (SOTA) methods, including traditional algorithms, such as vertex component analysis (VCA) [48], [49] and dynamic optimization-based EBE (DOEBE) [50], as well as six advanced DL networks: minimum simplex convolutional network (MiSiCNet) [51], deep autoencoder network (DAEN) [52], PGMSU [37], DGMSSU [38], SSAF-Net [39], and EOT-Net [43].

A. Datasets Introduction

As shown in Fig. 4, the experiments use three real datasets. Details are as follows.

1) *Samson Dataset:* Fig. 4(a) shows the Samson dataset [53]. Its size is 95×95 pixels with 156 bands. It covers the 401–889 nm wavelength range. It contains three endmembers: soil, tree, and water.
2) *Jasper Ridge Dataset:* Fig. 4(b) shows the Jasper Ridge dataset [53]. Its size is 100×100 pixels with 198 bands. It contains four endmembers: tree, water, dirt, and road.
3) *Urban Dataset:* Fig. 4(c) shows the Urban dataset. Its size is 307×307 pixels with 162 bands. It contains five endmembers: asphalt, grass, tree, roof, and dirt.

B. Experimental Setup

1) *Implementation Details:* The proposed PGT-Net is implemented in PyTorch using a full-image input strategy. We use the Adam optimizer with weight decay. The learning rate decays by a factor of 0.9 every 20 epochs. In the PGET module for interclass similarity modeling, the shared Transformer encoder consists of two layers with $h = 4$ linear attention heads and an embedding dimension D.

TABLE I
Network Parameter Value Setting

Dataset	Learning rate	Weight decay	Epochs	α	β	D
Samson	3e-4	2e-3	900	0.1	4.9	24
Jasper Ridge	5e-3	4e-5	700	0.8	0.2	32
Urban	3e-3	4e-5	200	0.8	0.2	40

As defined in the loss function, hyperparameters α and β control the relative contributions of the image mse and SAD loss. Their values are set according to the scene characteristics of different datasets and the roles of the two loss terms during optimization, as shown in Table I. For the Samson dataset, although the number of endmember classes is relatively small, the water endmember is easily affected by spectrally similar background components, thus requiring stronger spectral-shape discrimination. Therefore, under the fixed training setting, $\beta = 4.9$ is set as a larger weight to strengthen the SAD constraint and balance the mse and SAD losses, thereby improving interclass separability. On the contrary, Jasper Ridge and Urban contain more endmember classes and exhibit more complex mixed-pixel distributions, where stable amplitude reconstruction is more important for reliable abundance estimation. Accordingly, a smaller β is used so that the mse term plays a relatively larger role, maintaining image reconstruction accuracy while preserving spectral fidelity.

2) *Assessment Indicators:* We use SAD and root-mean-square error (RMSE) to quantitatively evaluate unmixing performance. The mathematical expressions are

$$\text{SAD} = \frac{1}{P}\sum_{i=1}^{P}\arccos\left(\frac{\hat{\mathbf{m}}_i^{\text{T}}\mathbf{m}_i}{\|\hat{\mathbf{m}}_i\|_2\|\mathbf{m}_i\|_2}\right) \tag{29}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{j=1}^{N}\|\hat{\mathbf{a}}_j - \mathbf{a}_j\|_2^2} \tag{30}$$

where $\hat{\mathbf{m}}_i$ and $\mathbf{m}_i$ are the extracted and reference endmember spectra, respectively. P is the number of endmembers. $\hat{\mathbf{a}}_j$ and $\mathbf{a}_j$ are the estimated and reference abundance vectors, respectively. N is the number of pixels. Lower SAD and RMSE values indicate better performance.

C. Experiment With Samson

Tables II and III report the quantitative results on the Samson dataset. Since VCA and DOEBE are endmember extraction methods, FCLS and SUnSAL were used to estimate abundances from their extracted endmembers. PGT-Net achieves the best overall performance, with the lowest average SAD and RMSE of 0.0641 and 0.0585. Although MiSiCNet and SSAF-Net obtain the best SAD values for the soil and

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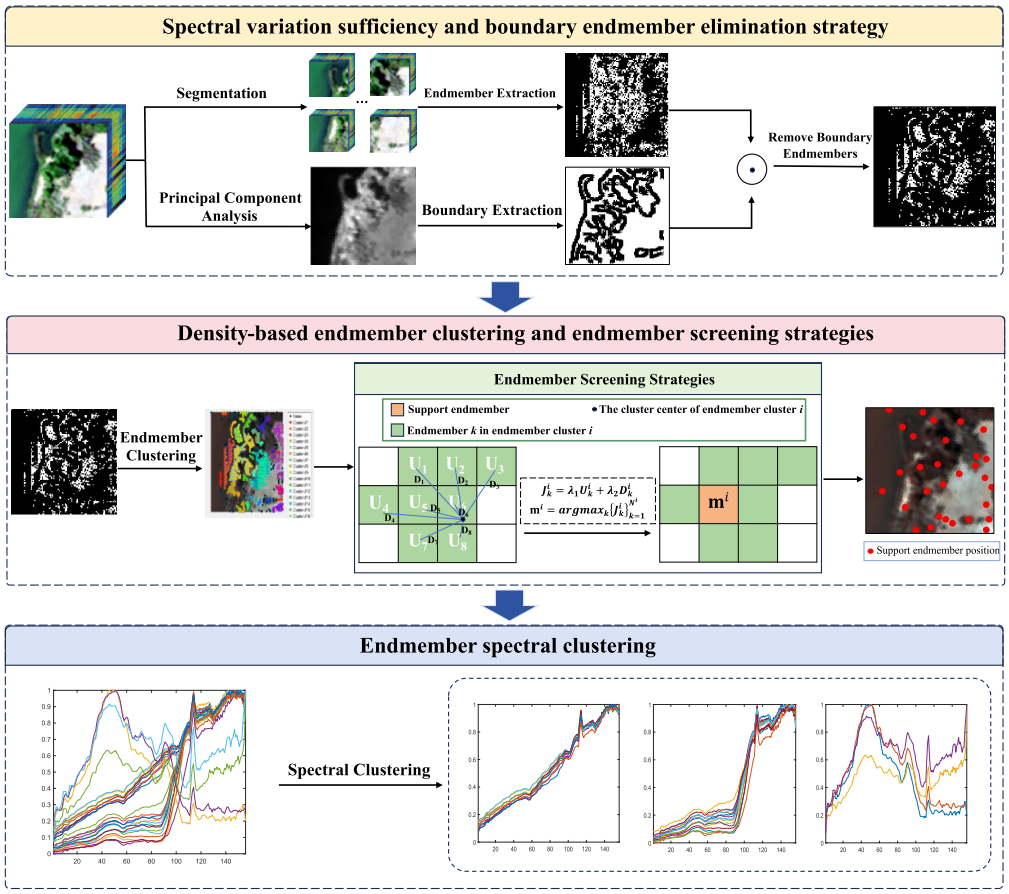


Fig. 2. Flowchart of the DR-EBE module.

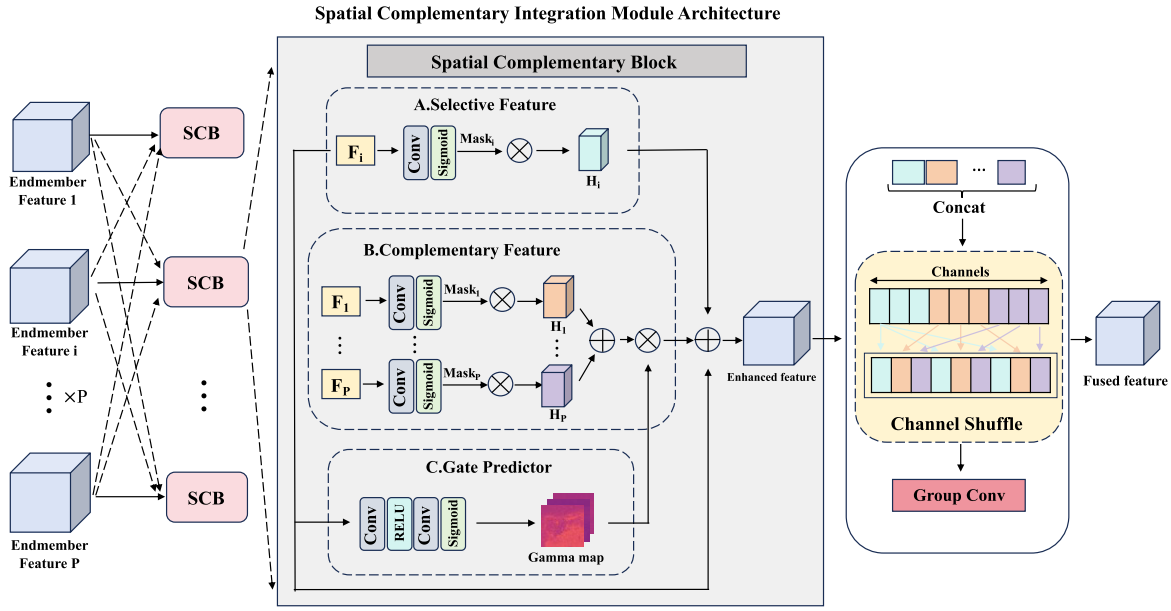


Fig. 3. Flowchart of the SCIM module, comprising spatial complementary fusion and channel interactive fusion.

tree endmembers, respectively, both show large spectral deviations for the water endmember. On the contrary, with the prompt-guided mechanism, PGT-Net enhances the discriminative features of easily confused endmembers and achieves the best results in both water endmember extraction and abundance estimation.

Figs. 5 and 6 visualize the extracted endmember spectra, abundance maps, and ground truth (GT). Overall, PGT-Net produces results closest to the GT. For the challenging water abundance estimation in the Samson dataset, most methods fail to accurately recover the high-abundance water region and its adjacent abundance-transition area, and tend to produce

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TABLE II
ENDMEMBER SAD RESULTS ON THE SAMSON DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	VCA	DOEBE	MiSiCNet	DAEN	PGMSU	DGMSSU	SSAF-Net	EOT-Net	**PGT-Net**
Soil	0.0236	0.0718	**0.0118**	0.1276	0.1291	0.0236	0.0222	0.0291	0.0286
Tree	0.0433	0.0458	0.0342	0.0538	0.0612	0.0867	**0.0292**	0.0481	0.0710
Water	0.1504	0.1901	0.3786	0.1555	0.1590	0.1211	0.4871	0.1586	**0.0927**
Mean SAD	0.0724	0.1025	0.1415	0.1123	0.1164	0.0771	0.1795	0.0786	**0.0641**

TABLE III
ABUNDANCE RMSE RESULTS ON THE SAMSON DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	VCA	DOEBE	MiSiCNet	DAEN	PGMSU	DGMSSU	SSAF-Net	EOT-Net	**PGT-Net**
Soil	0.2650	0.1206	0.1757	0.2249	0.2405	0.1849	0.1835	0.1082	**0.0704**
Tree	0.2511	0.0983	0.1819	0.1268	0.2212	0.2349	0.1716	0.0873	**0.0651**
Water	0.4235	0.1291	0.3077	0.1217	0.3414	0.3020	0.3063	0.0737	**0.0329**
Mean RMSE	0.3132	0.1160	0.2218	0.1578	0.2677	0.2406	0.2204	0.0909	**0.0585**

TABLE IV
ENDMEMBER SAD RESULTS ON THE JASPER RIDGE DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	VCA	DOEBE	MiSiCNet	DAEN	PGMSU	DGMSSU	SSAF-Net	EOT-Net	**PGT-Net**
Tree	0.1481	0.1188	0.0464	0.2061	0.0632	0.1112	0.0538	0.0458	**0.0370**
Water	0.2298	0.1565	0.2412	0.2283	0.1064	0.1925	0.2474	0.0724	**0.0505**
Dirt	0.1166	0.1004	0.0656	0.2293	0.1491	0.0334	0.0365	0.1070	**0.0277**
Road	0.0901	0.1889	0.3131	0.2615	0.3510	0.0362	0.0451	0.0629	**0.0325**
Mean SAD	0.1462	0.1411	0.1666	0.2313	0.1674	0.0933	0.0957	0.0720	**0.0369**

TABLE V
ABUNDANCE RMSE RESULTS ON THE JASPER RIDGE DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	VCA	DOEBE	MiSiCNet	DAEN	PGMSU	DGMSSU	SSAF-Net	EOT-Net	**PGT-Net**
Tree	0.1589	0.0849	0.0940	0.1362	0.1621	0.2031	0.0906	0.1299	**0.0659**
Water	0.2023	0.0734	0.1139	0.1169	0.1195	0.2592	0.1171	**0.0592**	0.0827
Dirt	0.1349	0.1188	0.2091	0.1607	0.1932	0.1827	0.1022	0.1769	**0.0805**
Road	0.1093	0.1378	0.2240	0.1713	0.1831	0.1185	0.0914	0.1144	**0.0730**
Mean RMSE	0.1514	0.1037	0.1603	0.1463	0.1645	0.1909	0.1003	0.1272	**0.0758**



(a) (b) (c)

Fig. 4. Dataset visualization. (a) Samson. (b) Jasper Ridge. (c) Urban.

erroneous water abundance estimates in nonwater background regions. On the contrary, PGT-Net extracts the water endmember more accurately and preserves a more coherent abundance distribution near the water boundary while effectively reducing background interference. These observations further verify the accuracy and reliability of the proposed method in both endmember extraction and abundance estimation.

D. Experiment With Jasper Ridge

Tables IV and V report the results on the more challenging Jasper Ridge dataset. PGT-Net again achieves the best overall performance. Compared with the second-best methods, PGT-Net reduces the mean SAD by 48.8% relative

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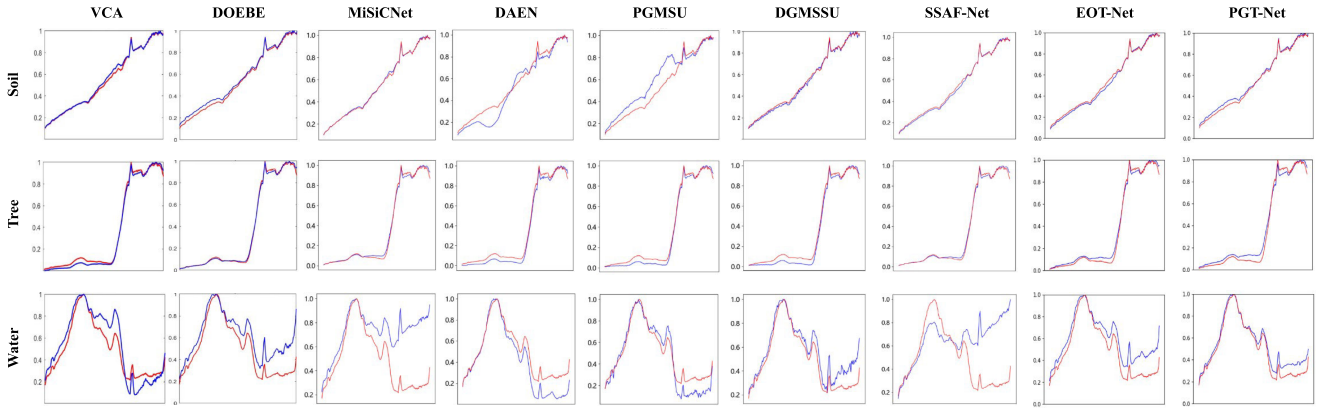


Fig. 5. Endmember spectra on the Samson dataset. Red lines denote the GT, and blue lines show the results extracted by various methods.

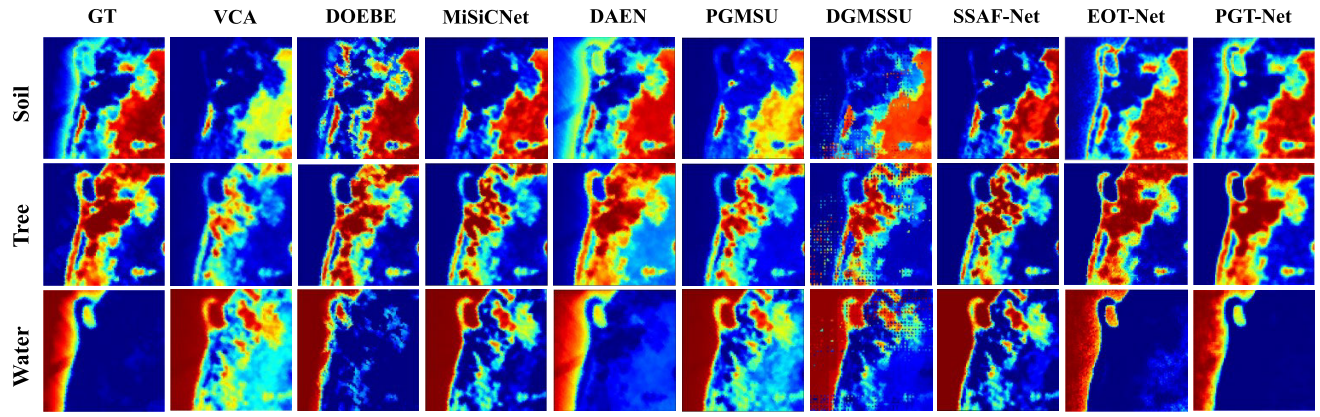


Fig. 6. Abundance maps on the Samson dataset.

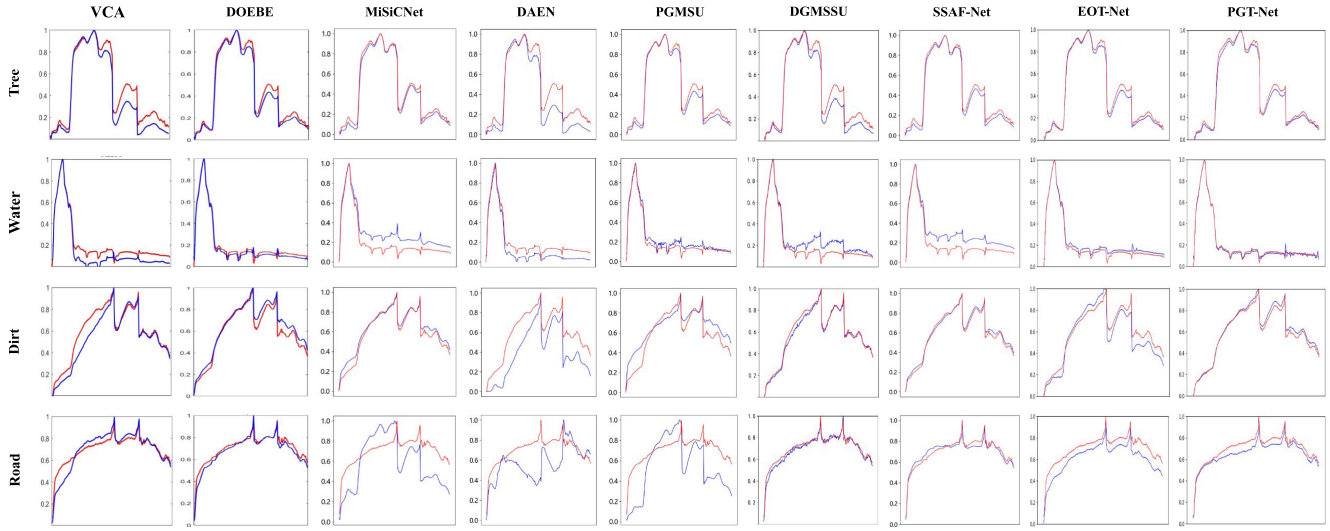


Fig. 7. Endmember spectra on the Jasper Ridge dataset. Red lines denote the GT, and blue lines show the results extracted by various methods.

to EOT-Net, reaching 0.0369, and reduces the mean RMSE by 24.4% relative to SSAF-Net, reaching 0.0758. The competing methods exhibit different limitations in endmember extraction and abundance estimation. Autoencoder-based methods, such as MiSiCNet and DAEN, show pronounced degradation on the road and water endmembers. Among generative models, PGMSU performs poorly on the road endmember. Although DGMSSU accurately extracts the dirt and road endmembers, this advantage does not translate into reliable abundance estimation, as indicated by its highest mean RMSE. SSAF-Net and EOT-Net achieve competitive overall results, but their performance remains uneven across individual end-

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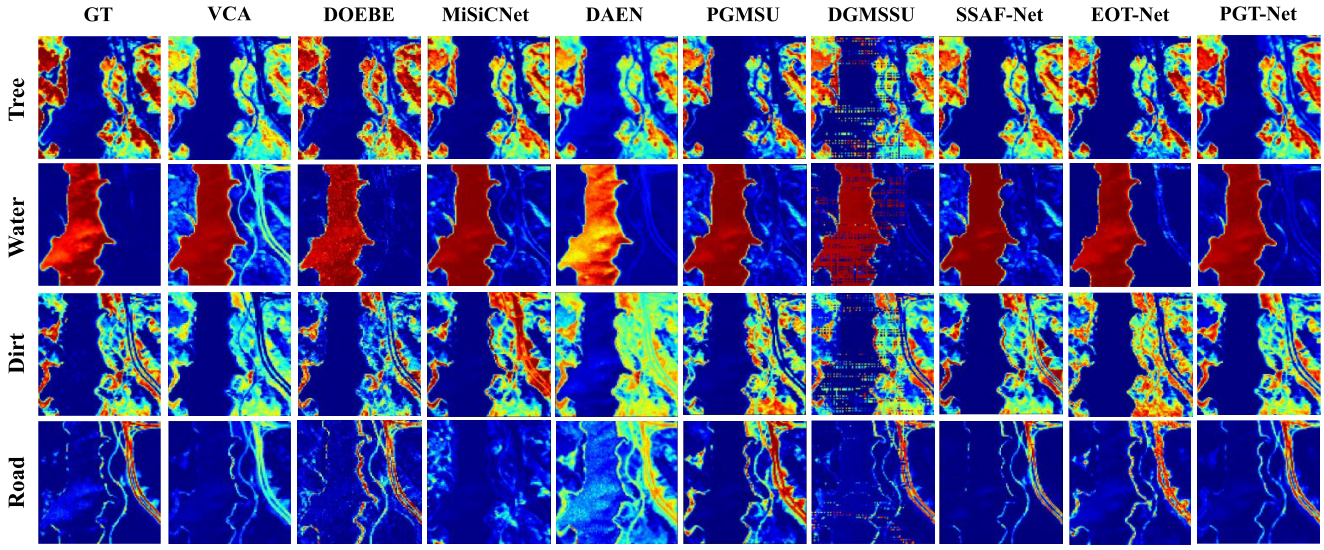


Fig. 8. Abundance maps on the Jasper Ridge dataset.

TABLE VI
ENDMEMBER SAD RESULTS ON THE URBAN DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	VCA	DOEBE	MiSiCNet	DAEN	PGMSU	DGMSSU	SSAF-Net	EOT-Net	**PGT-Net**
Asphalt	0.1416	0.0986	0.1391	0.1763	0.0974	0.0672	0.1338	0.0589	**0.0342**
Grass	0.1127	0.0773	0.3742	0.2391	0.2852	0.1991	0.2537	0.1128	**0.0299**
Tree	0.1103	0.1049	0.6979	0.2044	0.1761	0.0785	0.4295	**0.0708**	0.0891
Roof	0.1388	0.1018	**0.0521**	0.3718	0.0992	0.0615	0.1022	0.0552	0.0785
Dirt	1.1779	0.0777	0.4729	0.2311	0.1684	**0.0551**	0.2415	0.1318	0.0729
Mean SAD	0.3363	0.0921	0.3473	0.2445	0.1653	0.0923	0.2322	0.0859	**0.0609**

members. In particular, SSAF-Net shows limited spectral extraction for water, while EOT-Net yields a relatively large abundance error for dirt.

As shown in Figs. 7 and 8, PGT-Net generally produces results that most closely match the GT. Taking the road abundance map as a representative example, DAEN introduces noticeable background interference, and DGMSSU produces pronounced spatial artifacts. Although SSAF-Net achieves a competitive RMSE, its abundance map still attenuates some fine road structures. On the contrary, PGT-Net better preserves the continuity and overall contour of the elongated road structure while reducing spatial artifacts and structural discontinuities. These results demonstrate the strong capability of PGT-Net in handling complex boundaries and spatially heterogeneous mixed pixels.

E. Experiment With Urban

Tables VI and VII report the results on the more complex Urban dataset, where five endmembers are highly scattered and severely mixed. PGT-Net achieves the best overall quantitative performance. Compared with the second-best methods, it reduces the mean SAD from 0.0859 to 0.0609, achieving a 29.1% improvement over EOT-Net, and reduces the mean RMSE from 0.1437 to 0.1058, achieving a 26.4% improvement over DOEBE. This demonstrates its robustness in complex scenarios.

Furthermore, the high spectral similarity among urban materials makes asphalt and dirt particularly challenging to distinguish. As shown in Table VI and Fig. 9, most comparison methods exhibit noticeable spectral deviations or local fluctuations, especially for the asphalt endmember. On the contrary, PGT-Net produces smoother spectral curves that are more consistent with the GT. As illustrated in Fig. 10, the abundance maps of urban materials are highly fragmented, with complex spatial mixtures and many small structures. Several comparison methods suffer from background noise, misclassification, or blurred object boundaries, especially when extracting sparse targets such as roofs. By contrast, PGT-Net better preserves small targets and spatial contours, which is consistent with the RMSE results in Table VII.

F. Ablation Experiment

To evaluate the contribution of each module, ablation experiments were conducted on the three datasets to verify the effectiveness and synergy of the core modules in PGT-Net, as reported in Table VIII. Overall, the complete model achieves

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TABLE VII
ABUNDANCE RMSE RESULTS ON THE URBAN DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	VCA	DOEBE	MiSiCNet	DAEN	PGMSU	DGMSSU	SSAF-Net	EOT-Net	**PGT-Net**
Asphalt	0.3425	0.1559	0.3080	0.1982	0.2796	0.2923	0.3085	0.2564	**0.1196**
Grass	0.3153	0.1568	0.3433	0.2729	0.2572	0.3321	0.3453	0.2066	**0.1264**
Tree	0.2673	0.1390	0.2870	0.1558	0.1595	0.3356	0.3595	0.1225	**0.1085**
Roof	0.2258	0.1451	0.2783	0.1066	0.1182	0.1762	0.1539	0.1136	**0.0611**
Dirt	0.3712	0.1218	0.1320	0.1738	0.1414	0.1822	0.1801	0.2178	**0.1009**
Mean RMSE	0.3044	0.1437	0.2697	0.1815	0.1912	0.2637	0.2694	0.1917	**0.1058**

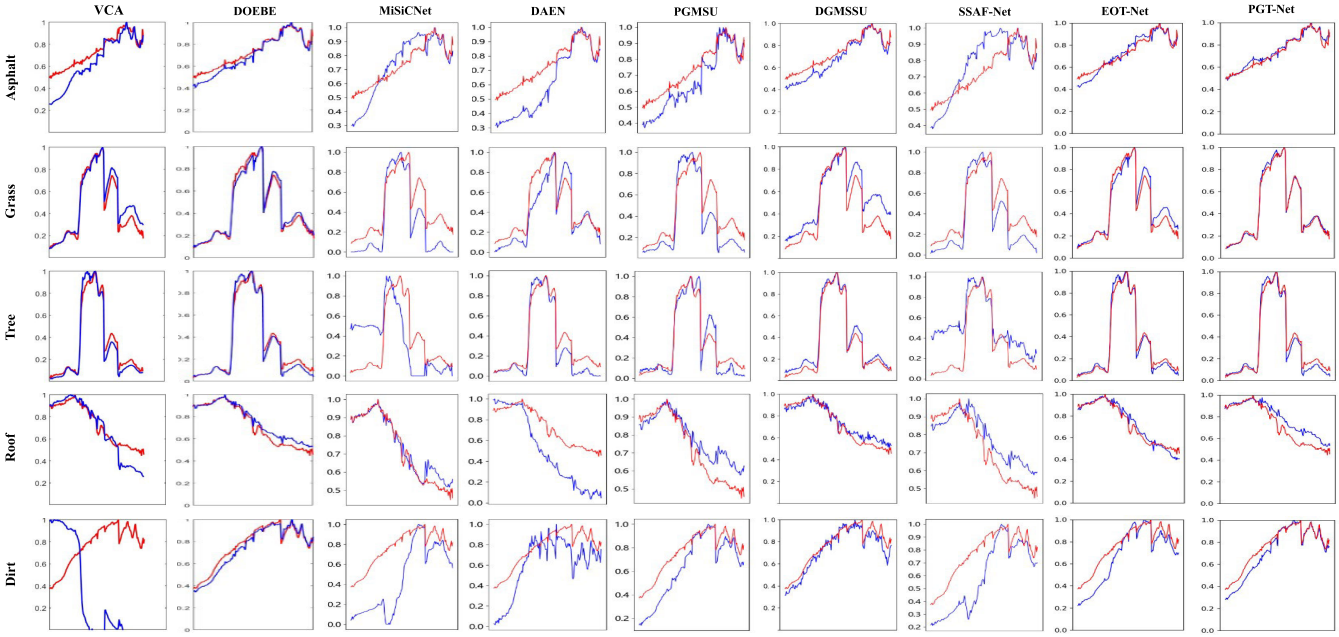


Fig. 9. Endmember spectra on the Urban dataset. Red lines denote the GT, and blue lines show the results extracted by various methods.

TABLE VIII
QUANTITATIVE RESULTS OF THE ABLATION STUDY. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Case	DR-EBE	PGET	SCIM	Samson		Jasper Ridge		Urban	
				SAD	RMSE	SAD	RMSE	SAD	RMSE
1	×	×	×	0.1052	0.1267	0.1189	0.1815	0.0859	0.1917
2	×	✓	✓	0.0938	0.0856	0.0837	0.1374	0.0781	0.1519
3	✓	×	✓	0.0824	0.0788	0.0690	0.1067	0.0724	0.1368
4	✓	✓	×	0.0703	0.0834	0.0458	0.1186	0.0652	0.1439
5	✓	✓	✓	**0.0641**	**0.0585**	**0.0369**	**0.0758**	**0.0609**	**0.1058**

the lowest SAD and RMSE on all three datasets, indicating that the three modules jointly improve the unmixing performance.

In particular, Case 1 removes all core modules and serves as the baseline network. It performs the worst on all datasets, showing that the basic network alone cannot sufficiently characterize SV in complex scenes. Comparing Case 2 with Case 5, removing DR-EBE leads to performance degradation, especially on the Jasper Ridge and Urban datasets, where the SAD increases from 0.0369 and 0.0609 to 0.0837 and 0.0781. This demonstrates that high-quality and low-redundancy endmember bundle priors are essential for subsequent prompt-guided targeted feature extraction. When endmember bundles still contain redundancy and noise, the network's ability to learn and distinguish endmember features is limited. Case 3 removes PGET. Although DR-EBE and SCIM enable it to outperform the baseline, its SAD remains much higher than that of the

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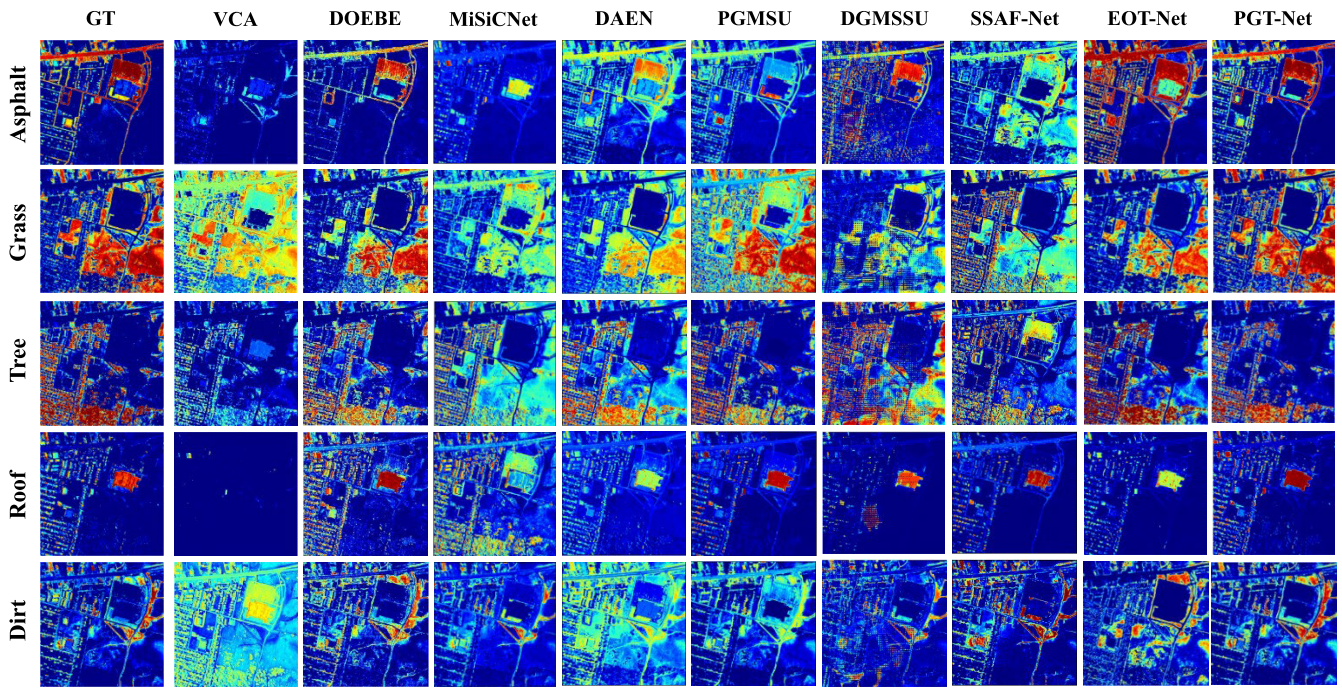


Fig. 10. Abundance maps on the Urban dataset.

complete model. This indicates that without prompt-guided activation, the model's targeted discrimination capability for highly similar endmembers is weakened, making subtle interclass differences in shared features difficult to fully activate and amplify. On the contrary, Case 4, which removes SCIM, still maintains a relatively low SAD, but both SAD and RMSE degraded compared with the complete model. The increase in RMSE is particularly evident: on the Jasper Ridge and Urban datasets, RMSE rises from 0.0758 and 0.1058 to 0.1186 and 0.1439. This suggests that SCIM improves abundance estimation accuracy through spatial complementary fusion and channel interaction. Finally, Case 5 achieves the best results on all datasets and metrics, further confirming that DR-EBE, PGET, and SCIM contribute from three complementary aspects: endmember prior quality, discriminative feature enhancement, and spatial abundance optimization. Their synergy therefore improves the overall unmixing performance.

V. Conclusion

To address large intraclass and small interclass SV, we propose PGT-Net, a prompt-guided Transformer unmixing network. It achieves high-precision unmixing through innovative targeted activation of endmember features and joint spatial-spectral constraints. First, the DR-EBE module employs a density-based spatial–spectral screening strategy to extract low-redundancy endmember bundles, accommodating intraclass variability and filtering noise. Next, the core PGET module converts endmember priors into endmember-specific prompt vectors via a prompt-guided mechanism. This accurately captures and amplifies interclass discriminative differences, overcoming feature confusion. Finally, the SCIM module leverages spatial complementarity for boundary refinement and cross-channel interaction, eliminating information isolation. The experimental results comprehensively demonstrate the effectiveness of PGT-Net. This work demonstrates the broad potential of the prompt-guided mechanism in HU. Future work will focus on developing more lightweight network architectures and further exploring adaptive endmember prompt construction and broader prompt learning applications for complex hyperspectral scenes.

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